Project Description: DUCSU eVote

DUCSU eVote is a web-based platform for running Dhaka University Central Students' Union (DUCSU) style elections online. The DUCSU election is one of the largest student elections in Bangladesh: in 2025, 471 candidates contested 28 central posts, and students voted on paper ballots in long queues, followed by many hours of manual counting.

DUCSU eVote aims to address this problem with a secure, transparent and privacy-preserving online voting system. Students log in with their registration number and a one-time password (OTP) sent to their university email, read candidate profiles and manifestos, compare panels, and cast their ballot from a phone or laptop.

The system is designed so that it records *that* a student has voted, but stores no data that shows *whom* they voted for. Voter records and ballots are kept in separate tables with no shared identifier, no timestamp and no ordering link between them. Each voter receives a receipt code that proves their ballot was included in the count without revealing their choices. A single database transaction with row locking ensures that each student can vote only once.

The platform will also include candidate and panel registration, an Election Commission dashboard, a read-only observer role, live turnout monitoring, automatic result calculation after voting closes, and a tamper-evident audit log of administrator actions.

The project will be developed as a fully deployable web application and demonstrated with dummy data; it is a prototype and is not intended to replace the official DUCSU election process. A student survey was conducted to identify real pain points, validate requirements and prioritise features. Its findings are summarised below.

Survey Summary

Survey Topic: Feasibility Study for DUCSU eVote

Method: Online questionnaire (Google Form), about 5 minutes, 27 questions in 5 sections

Respondents: 14 students of the University of Dhaka from 8 departments

The questionnaire covered five areas: respondent details; experience of the 2025 DUCSU election, with separate questions for voters and non-voters; trust and concerns about online voting; the importance of proposed features on a scale of 1 to 5; and open comments. Respondents were informed that their answers would be reported only as totals and percentages. Names and contact details of respondents are therefore not included in this report.

Experience of the 2025 Paper-Ballot Election

Twelve of the fourteen respondents (86%) voted in the 2025 DUCSU election; two did not. Voting on paper was slow for most of those who voted. Of the twelve who voted, seven took between one and two hours and one took more than two hours, including time spent waiting in line; one took 30 to 60 minutes and three finished within 30 minutes. Overall, two-thirds of voters (67%) spent more than an hour voting. They rated the ease of the overall process at an average of 3.0 out of 5, and only three of the twelve (25%) rated it 4 or 5.

The problems voters reported are shown in Table 1. A long queue was almost universal, and a lack of information about candidates was the second most common problem. More than half of the voters found the ballot confusing because of the large number of posts.

Table 1: Problems faced while voting (12 voters; multiple answers allowed)

Problem	Respondents	Percentage
Long queue	11	92%
Did not know enough about the candidates	8	67%
Ballot was confusing because of the many posts	7	58%
Hard to find polling centre or booth	6	50%
Worried that the vote was not secret	3	25%
Felt pressure from others about whom to vote for	3	25%
Missed classes or other work	3	25%

Both respondents who did not vote said they were busy with classes, exams or work, that the queue was too long, and that they live far from campus; one also said they were not interested. Both stated that they would have voted if they could have voted securely from their phone.

Willingness to Vote Online and Trust

Respondents were strongly willing to vote online. They rated how likely they would be to vote online at an average of 4.57 out of 5, with thirteen of fourteen (93%) choosing 4 or 5. Trust that an online

system would count votes correctly was similarly high (average 4.50; 93% rating 4 or 5).

Trust that an online system would keep the vote *secret* was lower, at an average of 4.07, with ten of fourteen (71%) rating it 4 or 5. Ballot secrecy is therefore the area where the system must work hardest to earn users' confidence. One respondent was strongly negative throughout, rating both likelihood of use and trust at 1 or 2 out of 5.

Concerns about Online Voting

Respondents could select up to three concerns. Their main worries were about the system being attacked or failing rather than about using it:

Table 2: Biggest worries about online voting (14 respondents; up to 3 answers each)

Concern	Respondents	Percentage
The system could be hacked	9	64%
The website could crash during voting	8	57%
Someone could find out whom I voted for	6	43%
Results could be changed by administrators	6	43%
I have no worries	3	21%
Someone could vote using my account	2	14%
People could be pressured or paid to vote at home	1	7%
Students without good internet would be left out	1	7%

Hacking and crashes were the most common worries. Fear that someone could find out how a respondent voted and fear that administrators could change results were each chosen by 43%. These concerns support the planned secret-ballot design, the read-only observer role and the tamper-evident audit log.

Measures that Would Increase Trust

When asked what would increase their trust in an online voting system, respondents most often chose open-source code that anyone can check (11 of 14, 79%) and a receipt code proving their vote was counted (10, 71%). These were followed by login with an OTP sent to the university email (8, 57%), live turnout during voting (7, 50%), independent observers (6, 43%), a public audit log of administrator actions (6, 43%) and results that can be checked against total votes (2, 14%). Open-source code, the most popular measure, was not part of the original proposal.

Feature Priorities

Every proposed feature received an average importance rating of at least 4.0 out of 5, so none should be removed. Based on average ratings, the five most important features are:

1. **Works Well on Mobile Phones** (4.57)
2. **Reviewing the Ballot before Submitting** (4.43)
3. **Vote Receipt Code** (4.29)
4. **Live Turnout during Voting** (4.21)
5. **Voting Reminders by Email or Notification** (4.21)

The remaining features were rated as follows: candidate profiles with photo and manifesto (4.14), results shown with charts after voting ends (4.07), Bangla language interface (4.07), and comparing two panels side by side (4.00).

Platform Preferences

All fourteen respondents use a smartphone with internet every day, and all of them (100%) preferred to log in with their registration number and an OTP sent to their university email. Thirteen (93%) would vote using a smartphone, and one would use a laptop or desktop computer. Twelve (86%) wanted the platform in both Bangla and English with an option to switch, and two preferred Bangla only; no respondent chose English only.

On whether candidate-wise vote counts should be visible while voting is still open, nine respondents (64%) said yes, three (21%) said no, and two (14%) were not sure. Three respondents (21%) agreed to a short follow-up interview.

Open Comments

Five respondents left written comments. One named the trustworthiness of system administrators as their biggest concern, and another asked whether the system can guarantee complete vote privacy and prevent hacking or manipulation of results. A third said there were so many candidates that they knew nothing about them. Two said they had no real concerns, one describing the project as a great initiative.

Implications for Requirements

The survey findings lead to the following decisions for DUCSU eVote:

1. **Mobile-first design.** 93% would vote by smartphone and mobile usability was the highest-rated feature (4.57), so every voter page will be designed and tested for phone screens first.
2. **Step-by-step ballot with review.** 58% of voters found the 28-post ballot confusing, and ballot review was rated 4.43. The ballot will show one post at a time with a progress indicator, followed by a full review screen before submission.
3. **Candidate information on the ballot.** Since 67% did not know enough about the candidates, each candidate on the ballot will link to their profile and manifesto, and the panel comparison tool will be kept.
4. **Security and reliability testing.** Because 64% fear hacking and 57% fear the website crash-

ing, security checks and load testing with a defined number of simultaneous voters become must-have non-functional requirements.

5. **Public source code.** 79% said open-source code would increase their trust, so the project source code will be published in a public repository, alongside the planned observer role and audit log.
6. **Explaining ballot secrecy.** Trust in secrecy (4.07) was lower than trust in counting (4.50), and 43% feared someone could find out how they voted, so a plain-language page will explain how the system keeps each vote secret.
7. **Voting reminders.** Reminders were rated 4.21, so the system will email voters when voting opens and before it closes.
8. **Bilingual interface.** With 86% wanting both languages, a Bangla/English switch becomes a should-have requirement for all voter-facing pages.
9. **Login method confirmed.** The unanimous preference for registration number with university email OTP confirms the planned authentication method.

Although 64% of respondents wanted candidate-wise vote counts to be visible during voting, the system will *not* show them. Early counts can influence students who have not yet voted and make it easier to pressure voters. Instead, live turnout, which respondents rated 4.21, will be shown during voting, and full results will be published as soon as voting closes. This decision will be explained to users on the voting page.

Feasibility Assessment

Operational feasibility. Demand is strong: 93% of respondents would be likely to vote online, both non-voters said they would vote from their phone, and all respondents use a smartphone daily.

Technical feasibility. All requested features can be built with standard web technologies (React, Node.js, PostgreSQL and email-based OTP) and require no special hardware.

Economic feasibility. Development and demonstration can be carried out on free hosting and email service tiers at no cost.

Schedule feasibility. The core voting system can be completed within the semester. Encrypted ballots will remain an optional extension, and the bilingual interface will be limited to voter-facing pages.

Legal and ethical feasibility. The system is feasible as a prototype using dummy data. Use in a real DUCSU election would require approval from the university and an independent security audit.

Conclusion

Respondents found paper voting slow, confusing and lacking in candidate information, and nearly all would vote online if the system were secure and reliable. Their main concerns, namely hacking, crashes, exposure of their vote and administrator tampering, are addressed by features already in the DUCSU eVote design and are strengthened by three additions from this survey: public source code, a page explaining ballot secrecy, and voting reminders. The survey confirms that DUCSU

eVote is feasible as a semester prototype.

Respondent Consent:

All respondents confirmed that they agreed to take part in the survey before answering. They were informed that their names would be used only to verify responses and that results would be reported only as totals and percentages.